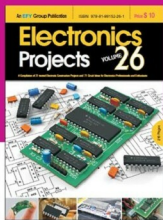


Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

It is a compilation of 21 construction projects and 71 circuit ideas published in Electronics For You magazine.



This collection of tested circuit ideas and construction projects in a handy volume would interest all classes of electronics enthusiasts—be they students, teachers, hobbyists or professionals!

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explode mask data volumes.

Another example is mid-range correction that becomes necessary for EUV masks because of wider dispersion of electrons during the mask-making process as these hit the multi-layer reflective mask.

Fab flow and variation

Patterning, CMP, deposition, etch and other process steps are challenging at 7nm and 5nm. Future devices would require structures with thin, precise and conformal films. And chipmakers would continue to grapple with structures that consist of only a finite number of atoms. It almost seems like a lot of technologies are all being worked on in parallel. Manufacturing with yield and having reliable devices would be difficult to do.

With those factors in mind, chipmakers face a sometimes-overlooked challenge, namely, process variation. Variation can be defined as any deviation from intended goal. There are various sources of variation in the fab, including within the die, within the wafer and between one tool chamber and another chamber. Variation control is really dropping down to atomic scale. Gate's critical dimension uniformity requirements are in angstroms not nanometers.

Selective processes

At current and future nodes, chipmakers would require newfangled technologies called selective deposition and selective removal. Combining novel chemistries with atomic-layer deposition or molecular-layer deposition tools, selective deposition involves a process of depositing materials and films in exact places.

Basically, most of our processing today is done based on line-of-sight capability. But soon, these would begin to tunnel through structures and build structures that you cannot see through from the top. So the industry needs to find a way to do selective deposition. Those tech-

niques are somewhat there, but are not mature enough.

Chipmakers, R&D organisations and universities are all working on selective deposition. Materials must also be removed using a related technology called selective removal or atomic-layer etch. This is a next-generation plasma etch technology that enables layer-by-layer, or atom-by-atom, etching. Atomic layer processing is the norm now for many applications.

Interconnects

In chip production, back-end-of-line is where interconnects are formed within a device. Interconnects and tiny wiring schemes in devices are becoming more compact at each node, thereby causing a resistance-capacitance delay in chips. Back-end-of-line would require new tools and materials. But if you make it taller, you may have more resistance-capacitance.

At 5nm, problems become worse. These occur due to the way wiring is done in the interconnect. It is not just in resistance and capacitance, but in how to wire up the transistor. We are seeing a lot of congestion. For that reason, the industry may need to consider an alternative path. This argument pushes us into doing 3D stacking.

Inspection and metrology

Optical inspection, the workhorse technology in the fab, is struggling to detect defects at 20nm and below. e-beam inspection can find tiny defects, but the technology is slow. Optical inspection tools operate at wavelengths down to 190nm. Seeking to capture more defects, a sub-190nm wavelength technology is used.

In addition, the industry is working on next-generation technology called multi-beam e-beam inspection. But this technology might not be ready until 2020. Exact timing will depend on how